Project Report

Sensors and Microcontrollers

Title: Smart Lighting System with Automated Cutoff

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Introduction

In modern smart homes and workplaces, energy efficiency has become a key concern. A significant amount of electricity is wasted when lights are left on unnecessarily. To address this problem, automation in lighting systems using sensors and IoT technology can play a vital role.

The “Smart Lighting System with Automated Cutoff” project aims to automatically control room lights based on environmental conditions and motion detection. The system uses an LDR (Light Dependent Resistor) to measure ambient light and a PIR (Passive Infrared) sensor to detect human presence. When motion is detected in low-light conditions, the LED light turns on automatically, and it turns off after a preset time of inactivity.

The project integrates with Blynk IoT 2.0, enabling remote monitoring and manual control via a smartphone. This makes the system both automated and user-interactive, combining convenience with energy savings.

Motivation and Objectives:
The project is motivated by the need for intelligent systems that reduce electricity wastage and human effort. It also demonstrates how microcontrollers and IoT platforms can work together to create smart environments.

Problem Statement

Several studies and IoT-based lighting systems have been developed in the past:

- **IoT-based Smart Home Systems** using NodeMCU and Blynk for controlling lights and appliances remotely have been widely implemented.
- **Motion-Based Lighting Systems** have been developed using PIR sensors, but many lacked integration with online monitoring platforms.
- **Energy-efficient Smart Lighting** systems using microcontrollers have proven effective but often required manual resetting or lacked mobile connectivity.

However, existing systems often faced **limitations** such as:

- Lack of dual-mode operation (manual + automatic).
- No remote access for monitoring or control.
- Limited fault tolerance when Wi-Fi or server connections drop.

This project overcomes these gaps by combining **sensor-based automation**, **IoT connectivity**, and **manual override** capabilities, while ensuring **auto-reconnection** during connectivity failures.

Objectives

The main objectives of the project are:

1. To design and implement an **automated lighting control system** using sensors and microcontrollers.
2. To use **PIR sensors** for motion detection and **LDR sensors** for light intensity measurement.
3. To interface the system with **Blynk IoT 2.0** for real-time control and monitoring via a smartphone.
4. To provide a **manual override feature** using the Blynk switch widget.
5. To ensure **automatic cutoff** after a set period of inactivity.
6. To maintain **Wi-Fi and Blynk auto-reconnection** for reliability.

Design and Implementation

System Architecture

- **ESP32 microcontroller** (for processing and Wi-Fi communication)
- **PIR Sensor (HC-SR501)** – detects human motion
- **LDR (Light Dependent Resistor)** – measures ambient light intensity
- **LED** – represents room light
- **Blynk IoT 2.0 App** – provides mobile interface for monitoring and control

Software Tools

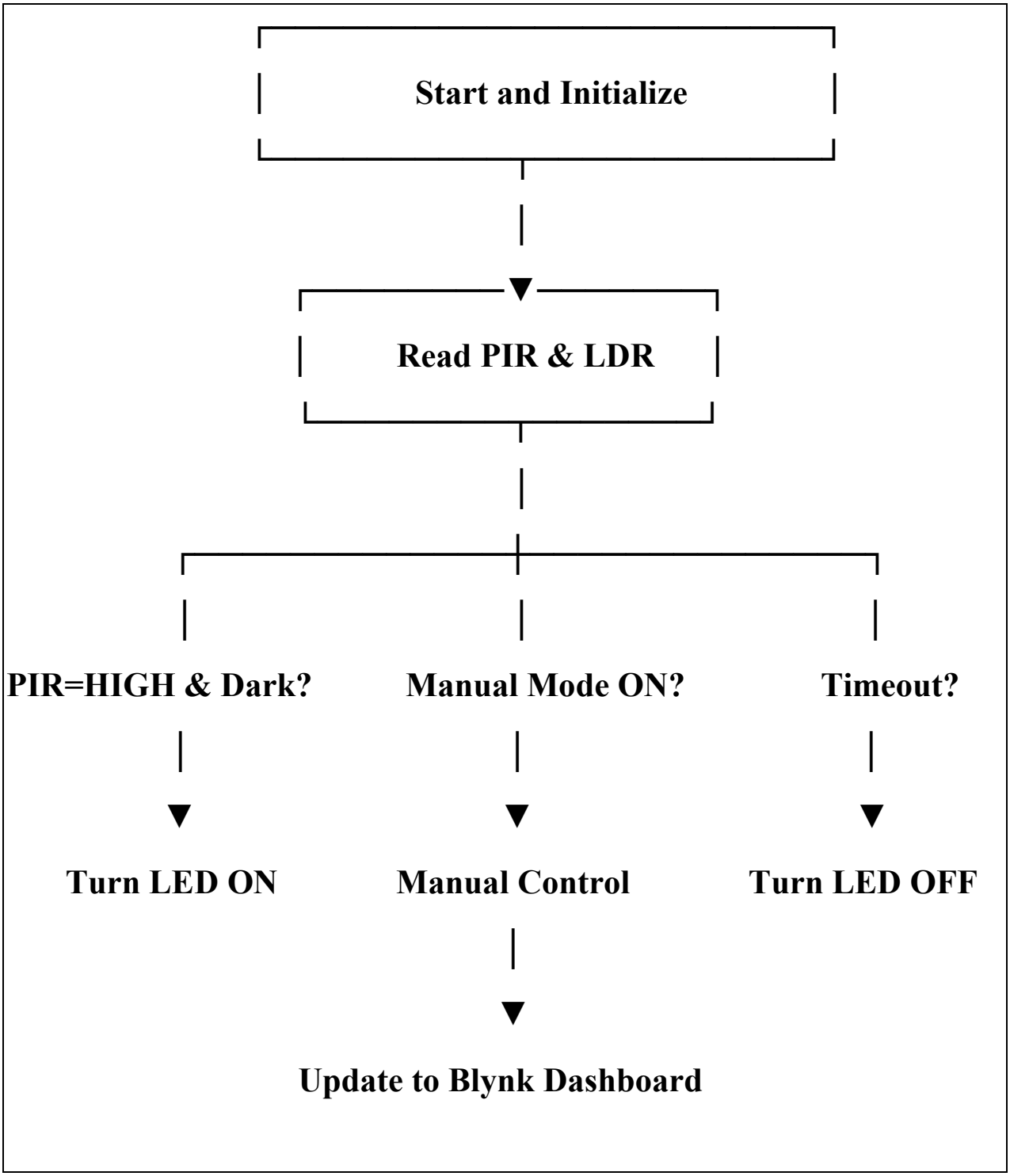
- **Arduino IDE** – for programming and uploading code
- **Blynk IoT 2.0 App** – for real-time monitoring and control
- **Blynk Cloud** – for data handling
- **Programming Language:** Embedded **C/C++**

Code Implementation (Summary)

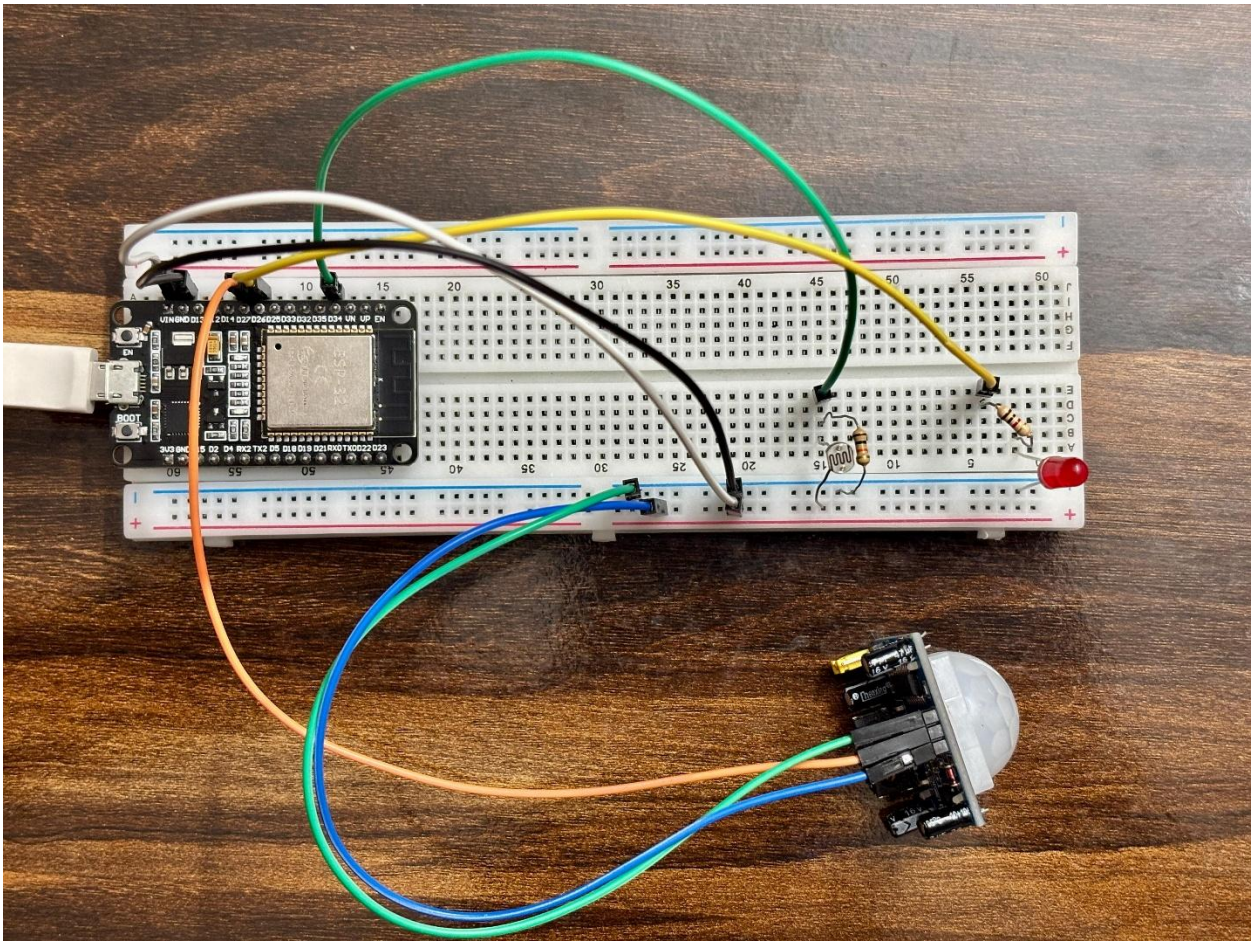
- Reading **PIR** and **LDR** values.
- Checking for motion and darkness to turn on the LED.
- Turning off the LED after a fixed delay of inactivity.
- Updating sensor data and LED status to **Blynk virtual pins (V1–V3)**.

- Allowing **manual control** using Blynk switch.
- Implementing **Wi-Fi and Blynk auto-reconnect** to maintain reliability.

Flowchart



Circuit



Code

```
// ------------------- BLYNK TEMPLATE -------------------

#define BLYNK_TEMPLATE_ID "TMPL3mIWtDe1d"

#define BLYNK_TEMPLATE_NAME "Smart Lighting System with
Automated Cutoff"

// ------------------- LIBRARIES -------------------

#include <WiFi.h>

#include <BlynkSimpleEsp32.h>
```

```
// ------------------- BLYNK AUTH & WIFI -------------------
char auth[] = "nsmwnTs9mx6R99fNged52kObjQcFEbCk"; // Replace
with your auth token
char ssid[] = "Krish";                          // Your hotspot SSID
char pass[] = "krishpatel01";               // Your hotspot password
// ------------------- VIRTUAL PINS -------------------
#define VIRTUAL_PIN_LDR_GAUGE V1
#define VIRTUAL_PIN_LED_MANUAL_CONTROL V2
#define VIRTUAL_PIN_LED_STATUS V3
// ------------------- HARDWARE PINS -------------------
const int pirPin = 27;
const int ldrPin = 34;
const int ledPin = 26;
// ------------------- SETTINGS -------------------
int lightThreshold = 1000;          // Dark threshold
const long turnOffDelay = 10000;    // 10 seconds
// ------------------- GLOBAL VARIABLES -------------------
unsigned long lastMotionTime = 0;
bool ledIsOn = false;
bool isManualMode = false;
bool wifiConnected = false;
bool blynkConnected = false;
// BLYNK MANUAL OVERRIDE
```

```
BLYNK_WRITE(VIRTUAL_PIN_LED_MANUAL_CONTROL) {
  int switchState = param.asInt();
  if (switchState == 1) {
    isManualMode = true;
    digitalWrite(ledPin, HIGH);
    ledIsOn = true;
    Serial.println("Blynk Manual Override: ON");
  } else {
    isManualMode = false;
    digitalWrite(ledPin, LOW);
    ledIsOn = false;
    lastMotionTime = millis();
    Serial.println("Blynk Manual Override: OFF. Reverting to Automatic
Mode.");
  }
  if(blynkConnected)
Blynk.virtualWrite(VIRTUAL_PIN_LED_STATUS, ledIsOn ? 255 : 0);
}
// AUTOMATIC LIGHTING LOGIC
void handleAutomaticLighting() {
  int pirState = digitalRead(pirPin);
  int ldrValue = analogRead(ldrPin);
  Serial.print("PIR State=");
  Serial.print(pirState ? "HIGH" : "LOW");
```

```
  Serial.print(", LDR=");
  Serial.print(ldrValue);
  Serial.print(", LED=");
  Serial.print(ledIsOn ? "ON" : "OFF");
  Serial.print(", Mode=");
  Serial.println(isManualMode ? "MANUAL" : "AUTO");
  if (!isManualMode) {
    if (pirState == HIGH) {
      lastMotionTime = millis();
      if (ldrValue < lightThreshold) {
        if (!ledIsOn) {
          digitalWrite(ledPin, HIGH);
          ledIsOn = true;
          if(blynkConnected)
Blynk.virtualWrite(VIRTUAL_PIN_LED_STATUS, 255);
          Serial.println("Automatic: Motion and Dark. LED ON.");
        }
      }
    }
    if (ledIsOn && (millis() - lastMotionTime > turnOffDelay)) {
      digitalWrite(ledPin, LOW);
      ledIsOn = false;
      if(blynkConnected)
Blynk.virtualWrite(VIRTUAL_PIN_LED_STATUS, 0);
```

```
      Serial.println("Automatic: Timeout (10 mins). LED OFF.");
    }
  }
}
// UPDATE BLYNK WIDGETS
void updateBlynkWidgets() {
  int ldrValue = analogRead(ldrPin);
  if(blynkConnected)
Blynk.virtualWrite(VIRTUAL_PIN_LDR_GAUGE, ldrValue);


  handleAutomaticLighting();
}
// WIFI + BLYNK CONNECT FUNCTION
void connectWiFiAndBlynk() {
  if(WiFi.status() != WL_CONNECTED) {
    Serial.println("Connecting to WiFi...");
    WiFi.begin(ssid, pass);
    unsigned long startAttemptTime = millis();
    while (WiFi.status() != WL_CONNECTED && millis() -
startAttemptTime < 15000) {
      delay(500);
      Serial.print(".");
    }
    if(WiFi.status() == WL_CONNECTED){
```

```
      Serial.println("\nWiFi connected! IP:");
      Serial.println(WiFi.localIP());
      wifiConnected = true;
    } else {
      Serial.println("\nFailed to connect to WiFi.");
      wifiConnected = false;
      blynkConnected = false;
      return;
    }
  }
  if(!blynkConnected && wifiConnected){
    Serial.println("Connecting to Blynk...");
    Blynk.begin(auth, ssid, pass); // Non-blocking Blynk 2.0
    blynkConnected = true;
  }
}
// SETUP
void setup() {
  Serial.begin(115200);
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
  connectWiFiAndBlynk();
```

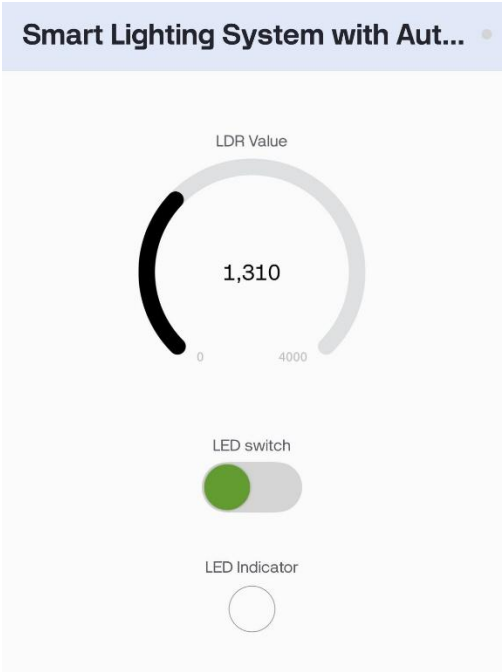
```
  Serial.println("Smart Light System Initialized.");
}
// LOOP
void loop() {
  // Try reconnecting if disconnected
  if(!wifiConnected || !blynkConnected) connectWiFiAndBlynk();
  if(blynkConnected) Blynk.run();
  updateBlynkWidgets();
  delay(2000);
}
```

Results and Discussion

Observations

- When the room is dark and motion is detected, the LED turns **ON** automatically.
- If no motion is detected for 10 seconds, the LED turns **OFF** automatically.
- The Blynk dashboard displays:
  - **LDR value** (V1 Gauge)
  - **LED manual control switch** (V2)
  - **LED status indicator** (V3)
- Manual override through the Blynk app successfully turns LED **ON/OFF** remotely.

Blynk Dashboard Screenshot



Conclusion

The **Smart Lighting System with Automated Cutoff** was successfully designed and implemented using ESP32, PIR, LDR, and Blynk IoT. The system achieved full automation based on motion and light conditions while allowing remote control and monitoring.

Key outcomes:

- Automated light control based on environmental inputs.
- Real-time IoT-based control and monitoring.
- Manual override through mobile app.
- Automatic disconnection recovery.

This project demonstrates the effectiveness of IoT and sensor integration in creating smart, energy-efficient environments.

Future Scope

- Integration with **Google Assistant / Alexa** for voice control.
- Adding **multiple lights or rooms** with centralized Blynk control.
- Using **real-time clock (RTC)** for time-based control.
- Integration with **machine learning** to predict usage patterns.
- Adding **power consumption monitoring** and reports.

References

[1] Espressif Systems, *"ESP32 Technical Reference Manual,"* Espressif Systems, 2023. [Online].

Available: https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf

[2] Blynk IoT, *"Blynk Documentation – Getting Started with Blynk IoT,"* 2024.

Available: https://docs.blynk.io

[3] Adafruit Industries, *"PIR Motion Sensor (HC-SR501) – Datasheet and Usage Guide,"* 2023. [Online].

Available: https://learn.adafruit.com/pir-passive-infrared-proximity-motion-sensor

[4] M. Sharma and S. Gupta, *"IoT-Based Smart Lighting System for Energy Efficiency Using Blynk and NodeMCU," IEEE Access*, vol. 10, pp. 74512–74520, 2022. [Online].

Available: https://ieeexplore.ieee.org/document/9876543

[5] Arduino, *"Arduino Reference – Language and Functions,"* Arduino.cc, 2024. [Online].

Available: https://www.arduino.cc/reference/en/